Rituraj Singh

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Education

National Institute of Technology Jamshedpur *2023-2027 (Expected)*
B.Tech in Electronics and Communication Engineering | CGPA: 8.16
Central Board of Secondary Education *2021-2023*
Class 12th | D.A.V Public School | Percentage: 75.4

Projects

- **Glimpse – A full-stack visual-sharing web app**
  React.js, Vite, JavaScript, Node.js, Express.js, MongoDB, Google Gemini API ***View Project***
  - Built a full-stack **visual-sharing platform** using a **React.js + Vite frontend** and a **MERN-based backend**, enabling users to create, explore, save, and interact with image-based content.
  - Implemented core features including **Google authentication with guest access control**, **image upload and pin management**, **search with category filtering**, and **commenting**, improving content discovery and engagement.
  - Integrated a **Google Gemini API-powered chatbot** to assist users with pin titles, category suggestions, profile improvement ideas, and content planning inside the application.
  - Designed a **responsive Pinterest-inspired UI** with Tailwind CSS and React Router, while organizing the codebase with reusable components, API utilities, and modular backend controllers for better scalability.
- **Redix – A lightweight Redis clone built in C++**
  C++, TCP Sockets, Multithreading, Ubuntu ***View Project***
  - Designed and developed a **Redis-inspired in-memory key–value store** in **C++**, supporting core commands with millisecond-level response time.
  - Implemented a **TCP server using sockets** with RESP protocol parsing to handle multiple concurrent client connections using multithreading and synchronization primitives.
  - Added support for **Stream operations** (`XADD`, `XRANGE`) with blocking reads and time-based ID generation.
  - Built a **Leader–Follower replication model** with replication ID and offset tracking to synchronize data across replicas.
- **Smart Product Pricing Model – A Multimodal AI Solution for Price Prediction**
  Python, Pandas, Matplotlib, Word2Vec, Vision Transformer (ViT), LightGBM, PCA ***View Project***
  - Performed extensive **data cleaning and EDA** on 50k+ product records to handle missing values, outliers, and inconsistent units, ensuring high-quality input for modeling.
  - Developed a **multimodal model** combining **text (Word2Vec embeddings)** and **image features (Vision Transformer – ViT)** for accurate price prediction.
  - Fused visual and textual representations using **PCA** and optimized a **LightGBM regressor**, achieving high prediction accuracy and generalization.
  - Applied **log normalization, feature scaling**, and iterative evaluation to enhance model robustness and prevent overfitting.

Skills

- **Languages**: Java, C++, Python, JavaScript
- **Core Computer Science**: Data Structures and Algorithms, OOP, DBMS, Operating Systems
- **Web Development**: React.js, Node.js, HTML, CSS, Tailwind CSS, REST APIs
- **Tools & Technologies**: Git, Docker, SQL, WSL, Vite
- **Machine Learning**: NumPy, Pandas, Scikit-learn, TensorFlow, NLP, Deep Learning, RNN

Achievements

- **Competitive Programming-** Solved **600+ problems** across LeetCode, Codeforces, and GeeksforGeeks.
- **Gen AI Study Jam,GDSC-** Completed Generative AI program by **Google Developer Student Clubs**. ***View Badge***
- **Data Science Certification-** Completed Data Science course covering Python, ML and DL fundamentals. ***View Certificate***
- **JEE Main-** Secured **AIR 17,543** (General) with **98.48 percentile** among **1.2M+ candidates**.

Extracurriculars

- **Content Team Member – SECE, NIT Jamshedpur**
  - Played an active role in organizing events and sessions conducted by the Society of Electronics and Communication Engineering (SECE).
- **Volunteer – Sankalp, NIT Jamshedpur**
  - Mentored underprivileged children, providing guidance on academics and career planning.

Online Profiles

- LeetCode
- Codeforces